Numbers, Percents & Ratios

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the complex number system, where $i^2 = -1$, the expression $\frac{a+6i}{2-i}$ is equal to $4 + bi$ for some real numbers a and b. What is the value of $a + b$?

To find the value of $a + b$, first relate the two sides of the equation: $\frac{a+6i}{2-i} = 4 + bi$ Multiply both sides by $2 - i$: $a + 6i = (4 + bi)(2 - i)$ Expand the right-hand side using the FOIL method and the identity $i^2 = -1$: $(4 + bi)(2 - i) = 8 - 4i + 2bi - bi^2$ Because $i^2 = -1$, the term $-bi^2 = -b(-1) = +b$: $8 - 4i + 2bi + b = (8 + b) + (2b - 4)i$ Now equate the corresponding real and imaginary components from both sides of $a + 6i = (8 + b) + (2b - 4)i$: 1. Equating imaginary parts.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

On the real number line below, point A has coordinate -16 and point C has coordinate 24. Point B has coordinate k, where $-16 < k < 24$. If the ratio of the distance between A and B to the distance between B and C is $3 : 5$, what is the value of k?

First, find the total distance between points A and C: $AC = 24 - (-16) = 40$ The point B divides segment AC in the ratio $3 : 5$, meaning the segment is divided into $3 + 5 = 8$ equal parts. The length of each part is: $\frac{40}{8} = 5$ The distance from A to B corresponds to 3 parts: $AB = 3 \times 5 = 15$ Since B lies to the right of A, its coordinate k is: $k = -16 + 15 = -1$ Alternatively, set up an algebraic proportion using the coordinate distances: $\frac{k-(-16)}{24-k} = \frac{3}{5}$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

On the number line shown below, point A is located at -16 and point B is located at 24. Point P lies on the line segment connecting A and B such that the ratio of the length of $\overline{AP}$ to the length of $\overline{PB}$ is $3 : 2$. What is the coordinate of point P?

To find the coordinate of point P, follow these steps: 1. Find the total distance between points A and B: $AB = 24 - (-16) = 24 + 16 = 40$ 2. Determine the fraction of the total distance represented by $\overline{AP}$. The ratio of the length of $\overline{AP}$ to $\overline{PB}$ is $3 : 2$, meaning there are $3 + 2 = 5$ equal parts in total. Thus, $\overline{AP}$ accounts for $\frac{3}{5}$ of the total distance: $AP = \frac{3}{5} \times 40 = 24$ 3. Add this distance to the starting coordinate of point A.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the distribution of the number of siblings reported by the 25 students in a homeroom class. If 1 student is selected at random from the class, what is the probability that the selected student has at least 2 siblings?

To find the probability that a randomly chosen student has at least 2 siblings: 1. Find the total number of outcomes (students): $6 + 10 + 6 + 3 = 25$ 2. Find the number of favorable outcomes. The phrase "at least 2 siblings" means having 2 or more siblings (either 2 or 3 siblings): - Number of students with 2 siblings: 6 - Number of students with 3 siblings: 3 - Favorable outcomes: $6 + 3 = 9$ 3. Calculate the probability: $P(\text{at least 2 siblings}) = \frac{9}{25}$ Common Misconceptions.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the number of hours four students volunteered at a community center last week. What percent of the total volunteer hours logged by the four students was logged by Maya?

To find the percentage of total hours logged by Maya: 1. Find the total number of volunteer hours logged by all four students: $\text{Total hours} = 8 + 14 + 6 + 12 = 40$ 2. Determine the fraction of hours Maya contributed: $\frac{\text{Maya's hours}}{\text{Total hours}} = \frac{14}{40}$ 3. Simplify the fraction and convert to a percentage: $\frac{14}{40} = \frac{7}{20} = \frac{35}{100} = 35\%$

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The bar graph below shows the number of books read by 4 students—Student A, Student B, Student C, and Student D—during summer break. Student A read 3 books, Student B read 5 books, Student C read 8 books, and Student D read 12 books. What is the average (arithmetic mean) number of books read by the 4 students?

To find the average (arithmetic mean) number of books read by the 4 students, find the sum of all books read and divide by the total number of students (4): $\text{Mean} = \frac{3 + 5 + 8 + 12}{4} = \frac{28}{4} = 7.0$

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The graph of the quadratic function $f(x) = -x^2 + bx + c$ is shown in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(1, k)$, and the graph passes through the point $(4, 0)$. What is the value of k?

We can solve for k using two algebraic methods: Method 1: Symmetry and Factored Form The axis of symmetry of the parabola passes vertically through the vertex, so its equation is $x = 1$. Since $(4, 0)$ is an x-intercept, its horizontal distance from the axis of symmetry is $4 - 1 = 3$. By symmetry, the other x-intercept must lie 3 units to the left of $x = 1$, which is at $x = 1 - 3 = -2$. Since the leading coefficient of $f(x) = -x^2 + bx + c$ is $a = -1$, we can write $f(x)$ in factored form.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with its vertex at $(3, 2)$ and passing through the point $(1, 6)$. What is the value of b?

1. Write the quadratic function in vertex form: $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. 2. Given the vertex $(3, 2)$, the equation becomes: $f(x) = a(x - 3)^2 + 2$ 3. Use the given point $(1, 6)$ to solve for the parameter a: $6 = a(1 - 3)^2 + 2$ $6 = a(-2)^2 + 2$ $6 = 4a + 2 \Rightarrow 4a = 4 \Rightarrow a = 1$ 4. Substitute $a = 1$ into the vertex form and expand to standard form $f(x) = ax^2 + bx + c$:
$f(x) = 1(x - 3)^2 + 2 = (x^2 - 6x + 9) + 2 = x^2 - 6x + 11$ 5. Comparing coefficients with.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the right triangle ΔABC shown below, the right angle is at vertex C. The lengths of the sides are given in inches: $AC = 12$, $BC = 5$, and $AB = 13$. What is the value of $\cos B$?

In any right triangle, the cosine of an acute angle is defined as the ratio of the length of the side adjacent to the angle to the length of the hypotenuse: $\cos B = \frac{\text{length of side adjacent to } B}{\text{length of hypotenuse}} = \frac{BC}{AB}$ From the given triangle: - The side adjacent to $\angle B$ is $BC = 5$. - The hypotenuse is $AB = 13$. Therefore, $\cos B = \frac{5}{13}$. Common Misconceptions: - $\frac{12}{13}$ represents.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC shown below, the measure of $\angle A$ is 65° and the measure of $\angle B$ is 45°. What is the value of x ?

According to the Triangle Angle Sum Theorem, the sum of the interior angle measures of any triangle is 180°. For ΔABC: $m\angle A + m\angle B + m\angle C = 180^\circ$ Substitute the given values:
$65^\circ + 45^\circ + x^\circ = 180^\circ$ $110^\circ + x^\circ = 180^\circ$ $x = 180 - 110 = 70$ Common Misconceptions: - 110 results from merely calculating the sum $65 + 45$ without subtracting from 180°. - 65 results from incorrectly assuming the triangle is isosceles with.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -4)$ and y-intercept $(0, -2)$. If the point $(6, c)$ lies on the graph, what is the value of c?

1. Identify the vertex form of the parabola: Since the vertex is $(h, k) = (2, -4)$, the function is:
$f(x) = a(x - 2)^2 - 4$ 2. Solve for the leading parameter a using the given y-intercept $(0, -2)$:
$-2 = a(0 - 2)^2 - 4$ $-2 = 4a - 4$ $2 = 4a \Rightarrow a = \frac{1}{2}$ 3. Determine the full equation of the function:
$f(x) = \frac{1}{2}(x - 2)^2 - 4$ 4. Find c by substituting $x = 6$: $c = f(6) = \frac{1}{2}(6 - 2)^2 - 4 = \frac{1}{2}(16) - 4 = 8 - 4 = 4$
Misconception traps: - Assuming $a = 1$ leads to.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has legs 9 and 12. Find the hypotenuse.

Apply the Pythagorean Theorem $a^2 + b^2 = c^2$: $c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = \mathbf{15}$ ACT Scaling Shortcut: Notice that both legs are multiples of 3: $9 = 3 \times 3$ and $12 = 3 \times 4$. This is a $3 - 4 - 5$ Pythagorean triple scaled by factor 3: $3 \times 5 = \mathbf{15}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A triangle has angles 50° and 60°. Find the third.

By the Triangle Angle Sum Theorem, the interior angles of any planar triangle sum to 180°: $\theta_1 + \theta_2 + \theta_3 = 180^\circ\ 50^\circ + 60^\circ + \theta_3 = 180^\circ \Rightarrow 110^\circ + \theta_3 = 180^\circ\ \theta_3 = 180^\circ - 110^\circ = \mathbf{70}^\circ$

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the area of a circle with radius 5.

Apply the circle area formula $A = \pi r^2$: $A = \pi(5)^2 = \mathbf{25\pi} \approx 78.54$ Trap Warning: Do NOT confuse area (πr^2) with circumference ($2\pi r = 10\pi$). Most ACT answer choices leave π exact.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

The quadratic function $f(x) = x^2 - 2kx + (3k + 10)$ has 2 distinct real zeros. If both zeros are strictly greater than 2, how many integer values of k are possible?

To have both zeros of the upward-opening parabola $f(x) = x^2 - 2kx + (3k + 10)$ be real, distinct, and strictly greater than 2, three conditions must hold simultaneously: 1. Positive Discriminant ($\Delta > 0$): $\Delta = (-2k)^2 - 4(1)(3k + 10) = 4k^2 - 12k - 40 = 4(k - 5)(k + 2) > 0$. This requires $k > 5$ or $k < -2$. 2. Vertex Position ($x_v > 2$): The axis of symmetry of $f(x)$ is at $x = -\frac{-2k}{2(1)} = k$. For both zeros to lie to the right of 2, the midpoint of the zeros must satisfy: $k > 2$. Intersec.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

The graph of the quadratic function $f(x) = 2x^2 - 12x + c$ in the standard (x, y) coordinate plane passes through the point $(1, -3)$. What is the minimum value of $f(x)$?

Step 1: Determine the unknown parameter c by substituting the coordinates of the given point $(1, -3)$ into the function equation: $f(1) = 2(1)^2 - 12(1) + c = -3$ $2 - 12 + c = -3$ $-10 + c = -3 \Rightarrow c = 7$ Step 2: Write the complete function: $f(x) = 2x^2 - 12x + 7$ Step 3: Find the x-coordinate of the vertex. For a parabola $y = ax^2 + bx + c$, the vertex occurs at $x = -\frac{b}{2a}$: $x = -\frac{-12}{2(2)} = \frac{12}{4} = 3$ Step 4: Find the minimum value by evaluating $f(3)$.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

In the figure below, ΔABC is a right triangle with right angle at C, and altitude $\overline{CD}$ of length h is perpendicular to hypotenuse $\overline{AB}$. If $AD = 4$ and $DB = 9$, what is the area of ΔABC?

To find the area of ΔABC, we need the base AB and the corresponding altitude $CD = h$. 1. Find the altitude h: By the Geometric Mean (Altitude) Theorem for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $h^2 = AD \cdot DB = 4 \cdot 9 = 36$ Taking the square root gives: $h = \sqrt{36} = 6$ 2. Find the total length of base $\overline{AB}$: $AB = AD + DB = 4 + 9 = 13$ 3. Calculate the area of ΔABC.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

If $\cos\theta = \frac{12}{13}$ and $0^\circ < \theta < 90^\circ$, what is the value of $\tan\theta$?

In a right triangle with acute angle θ, the cosine ratio is defined as: $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{12}{13}$ We can represent the adjacent leg as 12 and the hypotenuse as 13. Using the Pythagorean theorem to find the length of the opposite leg: $\text{opposite} = \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5$ The tangent ratio is defined as: $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{5}{12}$ - $\frac{5}{13}$ is incorrect because i.

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Trigonometry

In the figure below, ΔABC is a right triangle with the right angle at B, and $\overline{BD}$ is an altitude to hypotenuse $\overline{AC}$. The length of $\overline{AD}$ is 4 units and the length of $\overline{DC}$ is 9 units. What is the area, in square units, of ΔABC ?

To find the area of ΔABC, we need the base AC and the altitude BD. 1. Find the length of altitude $\overline{BD}$: By the Right-Triangle Altitude Theorem (Geometric Mean Theorem), the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $BD^2 = AD \cdot DC$
$BD^2 = 4 \cdot 9 = 36 \Rightarrow BD = \sqrt{36} = 6$ 2. Find the length of hypotenuse $\overline{AC}$: $AC = AD + DC = 4 + 9 = 13$ 3. Calculate the area of ΔABC.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Statistics & Probability

In right triangle ΔABC, the measure of $\angle C$ is 90°. The length of side AC is 12 and the length of side BC is 5. What is the value of $\cos A$?

In right triangle ΔABC with the right angle at C, side AC is the leg adjacent to $\angle A$, side BC is the leg opposite to $\angle A$, and side AB is the hypotenuse. First, apply the Pythagorean theorem to find the length of the hypotenuse AB: $AB = \sqrt{AC^2 + BC^2} = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$ By the definition of cosine in a right triangle: $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{AC}{AB} = \frac{12}{13}$ - Choosing $\frac{5}{13}$ confuses c.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the value of the numerical expression $|-14| - 3|5-9| + (-2)^3$?

a. -6 ✓

b. -10

c. 2

d. 14

$|-14| = 14$, $|5-9| = |-4| = 4$, and $(-2)^3 = -8$. Substituting gives $14 - 3(4) + (-8) = 14 - 12 - 8 = -6$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If p is a positive prime number and q is a negative integer, which of the following values must always be negative?

a. $p - q$

b. $q - p$ ✓

c. $p^2 + q^2$

d. $\frac{p}{q^2}$

Since $p > 0$ and $q < 0$, $-p < 0$. Therefore, $q - p$ is the sum of two negative numbers, which is always negative.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the greatest common factor of 72, 108, and 180?

a. 18

b. 24

c. 36 ✓

d. 72

Prime factorizations: $72 = 2^3 \times 3^2$, $108 = 2^2 \times 3^3$, $180 = 2^2 \times 3^2 \times 5$. GCF $= 2^2 \times 3^2 = 4 \times 9 = 36$.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

On a blueprint, a scale of $\frac{1}{4}$ inch represents 3 feet. If a conference room has a length of 36 feet, what is its length on the blueprint?

a. 2.5 inches

b. 3 **inches** ✓

c. 4.5 inches

d. 6 inches

Set up the proportion: $\frac{0.25 \text{ in}}{3 \text{ ft}} = \frac{x \text{ in}}{36 \text{ ft}}$. Cross-multiplying yields $3x = 0.25 \times 36 = 9 \Rightarrow x = 3$ inches.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A car travels 180 miles on 6 gallons of fuel. At the same rate of fuel consumption, how many gallons of fuel are required to travel 450 miles?

a. 12

b. 15 ✓

c. 16.5

d. 18

Fuel efficiency is $\frac{180}{6} = 30$ miles per gallon. For 450 miles: $\frac{450}{30} = 15$ gallons.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The ratio of copper to zinc in a brass alloy is $5 : 3$ by weight. If a piece of brass weighs 56 kilograms, how many kilograms of copper does it contain?

a. 21

b. 30

c. 35 ✓

d. 40

Total ratio parts $= 5 + 3 = 8$. Each part weighs $\frac{56}{8} = 7$ kg. Weight of copper $= 5 \times 7 = 35$ kg.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A store purchases a bicycle for 150 and marks up the price by 40%. During a weekend clearance sale, the bicycle is discounted by 20%. What is the final sale price?

a. 168 ✓

b. 174

c. 180

d. 192

Retail price after 40% markup $= 150 \times 1.40 = 210$. After 20% discount, final price $= 210 \times 0.80 = 168$.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If 35% of a number N is equal to 84, what is 60% of N?

a. 120

b. 144 ✓

c. 168

d. 240

$0.35N = 84 \Rightarrow N = \frac{84}{0.35} = 240$. Then 60% of $240 = 0.60 \times 240 = 144$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The price of an airline ticket increased from 250 to 320. What was the percent increase in the ticket price?

a. 22%

b. 25%

c. 28% ✓

d. 30%

Increase $= 320 - 250 = 70$. Percent increase $= \frac{70}{250} \times 100\% = \frac{7}{25} \times 100\% = 28\%$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the value of $(3.2 \times 10^7) \times (2.5 \times 10^{-3})$ expressed in standard scientific notation?

a. 8.0×10^4 ✓

b. 8.0×10^5

c. 8.0×10^3

d. 0.8×10^5

Multiply coefficients: $3.2 \times 2.5 = 8.0$. Add exponents: $10^{7+(-3)} = 10^4$. The result is 8.0×10^4.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following is equivalent to $\frac{x^5 \cdot x^{-2}}{(x^3)^2}$ for all positive values of x?

a. x^{-3} ✓

b. x^{-1}

c. x^0

d. x^2

Numerator $= x^{5-2} = x^3$. Denominator $= x^{3\times 2} = x^6$. Quotient $= \frac{x^3}{x^6} = x^{3-6} = x^{-3}$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the radical expression $\sqrt{75} - 2\sqrt{27} + \sqrt{48}$.

a. $2\sqrt{3}$

b. $3\sqrt{3}$ ✓

c. $4\sqrt{3}$

d. $5\sqrt{3}$

Simplify each radical: $\sqrt{75} = 5\sqrt{3}$, $2\sqrt{27} = 2(3\sqrt{3}) = 6\sqrt{3}$, and $\sqrt{48} = 4\sqrt{3}$. Combining yields $(5 - 6 + 4)\sqrt{3} = 3\sqrt{3}$.

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Every student in the graduating class of Oakridge High School plans to do exactly one of the following after graduation: attend a 2-year college, attend a 4-year university, or enter the workforce directly. • The ratio of the number of students planning to attend a 2-year college to the number of students planning to attend a 4-year university is $3:5$. • Of the students planning to attend a 2-year college, exactly 80% will attend an in-state college. • Of the students planning to attend a 4-year university, exactly 72% will attend an in-state university. • The total number of students planning to attend an in-state college or university accounts for exactly 45% of the entire graduating class. If the number of students in every category and subcategory must be a positive integer, what is the minimum possible number of students in the graduating class?

a. 40

b. 100

c. 120

d. **200** ✓

Let N be the total number of students in the graduating class. 1. **Express college counts in terms of a multiplier x:** The ratio of students attending a 2-year college to those attending a 4-year university is $3:5$. Thus, for some positive value x: $C_2 = 3x \quad \text{and} \quad C_4 = 5x$ Since both $3x$ and $5x$ must be integers, and $\gcd(3, 5) = 1$, $x = 2(5x) - 3(3x)$ must be a positive integer. 2. **Determine the number of in-state students:** - In-state 2-year college students.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which of the following is a prime number?

a. **43** ✓

b. 36

c. 38

d. 44

43 has no positive divisors other than 1 and itself, so it is prime; the other choices each factor into smaller integers.

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35 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A boutique owner purchases a designer jacket at a wholesale price of W dollars and marks up the price by $p\%$. During an end-of-season clearance sale, this marked-up price is discounted by $p\%$. The final clearance price is $x84lessthantheoriginalwholesaleprice, andtheclearancepriceis\336$ less than the marked-up price. What is the value of W, the original wholesale price?

a. \672

b. **\756 ✓**

c. \1,008

d. \1,344

Let $r = \frac{p}{100}$. The marked-up price, M, is given by: $M = W(1 + r)$ The clearance sale price, S, is discounted by $p\%$ from M: $S = M(1 - r) = W(1 + r)(1 - r) = W(1 - r^2)$ We are given two pieces of information: 1. The clearance price is $x84lessthanthewholesaleprice$: W - S = 84 \implies W - W(1 - r^2) = 84 \implies W r^2 = 842. $Theclearancepriceis\336$ less than the marked-up price:
$M - S = 336 \Rightarrow W(1 + r) - W(1 - r^2) = 336$ Expanding the left-hand side yields.

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

In a high school choir of 60 students, 40% are sopranos, 25% are altos, and the remaining students are tenors or basses. How many students in the choir are tenors or basses?

a. 15

b. **21 ✓**

c. 24

d. 39

First, find the percentage of students in the choir who are tenors or basses by subtracting the percentages of sopranos and altos from 100%: $100\% - (40\% + 25\%) = 100\% - 65\% = 35\%$ Next, calculate 35% of the total 60 choir students: $0.35 \times 60 = 21$ Therefore, there are 21 students who are tenors or basses. Common misconceptions: - 15 corresponds only to the number of altos (0.25×60). - 24 corresponds only to the number of sopranos (0.40×60). - 39 corresponds to the comb.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A high school track team has 45 members, and the ratio of sprinters to distance runners is 4 to 5. If every member is either a sprinter or a distance runner, how many members of the team are distance runners?

a. 20

b. 25 ✓

c. 30

d. 36

The ratio of sprinters to distance runners is $4:5$, meaning the team is divided into $4+5=9$ equal parts. To find the size of each part: $\frac{45}{9}=5$ Since distance runners represent 5 of these equal parts, the number of distance runners is: $5\times5=25$ Common errors: - 20 represents the number of sprinters (4×5). - 36 incorrectly takes $\frac{4}{5}\times45$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

How many positive factors does 72 have?

a. 15

b. 17

c. 7

d. **12** ✓

The positive factors of 72 are 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72 — a total of 12.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

After a 25\% markup, an item costs $x25$. What was the original price?

a. $x118$

b. $x182$

c. $x196$

d. x**20** ✓

Let original $=x$. $x(1+25/100)=25\Rightarrow x=20$. Original price: $x20$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Statistics & Probability

After a 10\% discount, an item costs $x63$. What was the original price?

a. $x151$

b. x**70** ✓

c. $x22$

d. $x86$

Let original $= x$. $x(1 - 10/100) = 63 \Rightarrow x = 70$. Original price: $x70$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

A total of 70 is divided in the ratio 2:5:7. Find the middle share.

a. 47

b. 65

c. 4

d. **25** ✓

Total parts $= 14$, each part $= 70/14 = 5$. Shares: 10, 25, 35. The middle is 25.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Elementary & Intermediate Algebra

A total of 48 is divided in the ratio 1:3:4. Find the smallest share.

a. 29

b. 40

c. **6** ✓

d. 36

Total parts $= 8$, each part $= 48/8 = 6$. Shares: 6, 18, 24. The smallest is 6.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Functions & Graphs

The sum of 9 consecutive integers is 342. What is the smallest of these integers?

a. **34** ✓

b. 23

c. 30

d. 43

Let the smallest be x. Then $9x + 36 = 342 \Rightarrow x = 34$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Coordinate Geometry

Split 12 in the ratio 1:3. Find the smaller part.

a. 11

b. 5

c. 7

d. **3** ✓

$1 + 3 = 4$ parts, each $= 12/4 = 3$. Smaller part $= 1 \times 3 = 3$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Plane Geometry

Split 15 in the ratio 1:2. Find the larger part.

a. **10** ✓

b. 2

c. 12

d. 4

$1 + 2 = 3$ parts, each $= 15/3 = 5$. Larger part $= 1 \times 5 = 10$.

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